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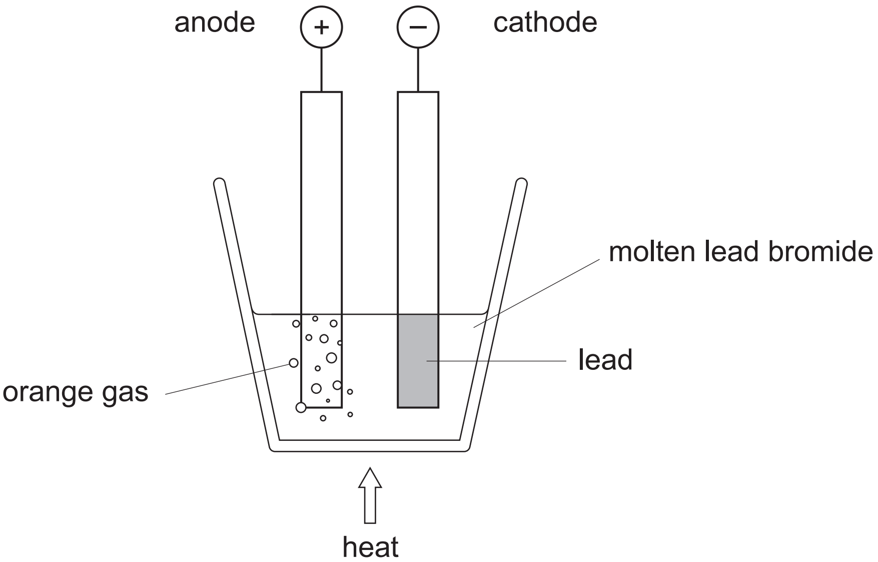
4. (a) The diagram shows the apparatus a teacher used to demonstrate what happens when an electric current is passed through molten lead bromide.



(i) Lead bromide contains the ions Pb^{2+} and Br^{-}.

<u>Underline</u> the correct formula of lead bromide. [1]

Pb_2Br **PbBr** **$PbBr^2$** **$PbBr_2$**

(ii) Give the state (solid, liquid or gas) of the lead bromide during the process. [1]

....

(iii) Name the orange gas formed at the anode. .... [1]

(iv) During the process each lead ion, Pb^{2+}, gains two electrons forming a lead atom at the cathode. [1]

Tick (✓) the box next to the equation for the reaction at the cathode.

$Pb + 2e^{-} \longrightarrow Pb^{2+}$	☐
$Pb^{2+} - 2e^{-} \longrightarrow Pb$	☐
$Pb^{2+} + 2e^{-} \longrightarrow Pb$	☐
$Pb - 2e^{-} \longrightarrow Pb^{2+}$	☐

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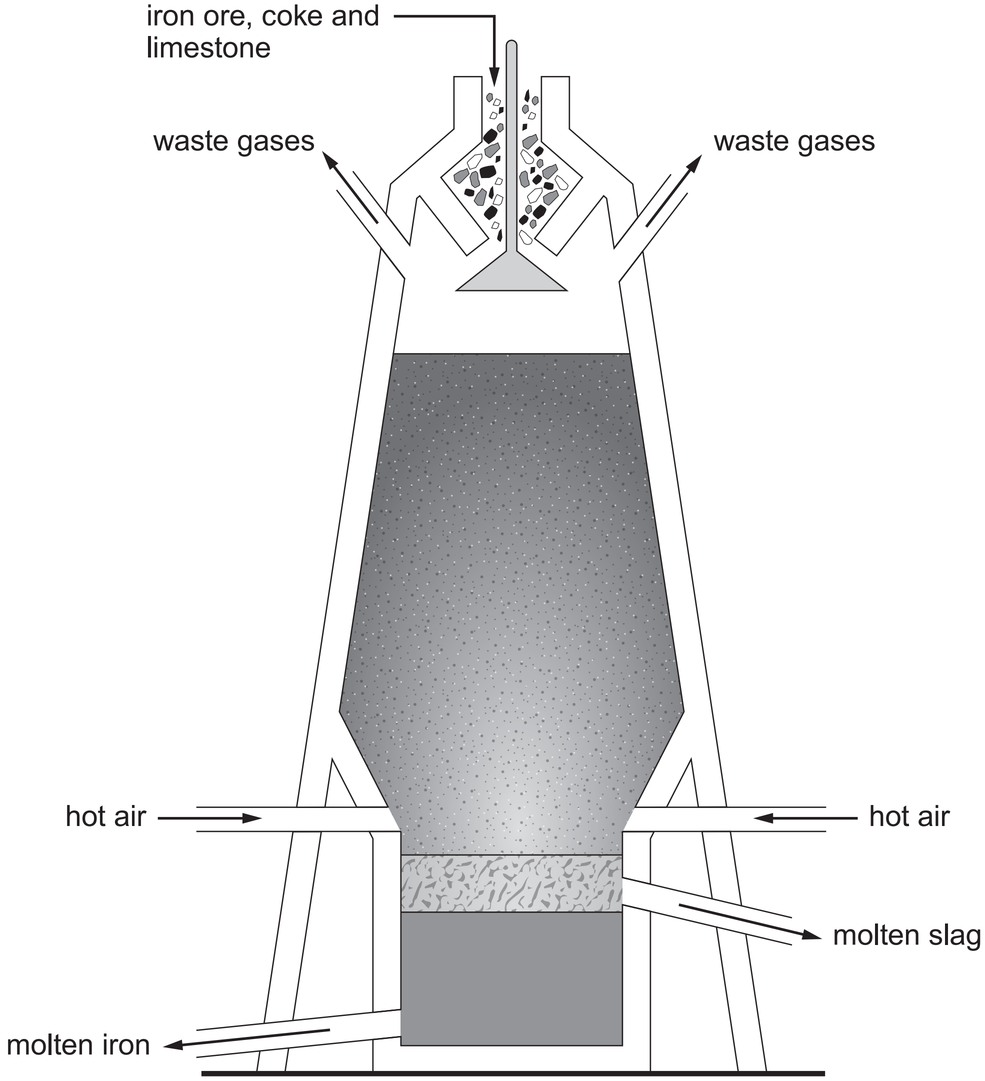
(b) Iron is extracted from its ore in the blast furnace. The diagram shows the materials which enter and leave the furnace.



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(i) Underline the correct word(s) in the brackets to complete each sentence. [3]

The furnace is heated by burning (**coke** / **iron** / **iron ore**).

Hot air provides (**waste gases** / **oxygen** / **slag**) for burning to take place.

Impurities are removed by adding (**steel** / **hot air** / **limestone**) to the furnace.

(ii) The word equation shows the main reaction taking place in the blast furnace.

iron(III) oxide + carbon monoxide ⟶ iron + carbon dioxide

Give the **letter** next to the products which correctly balance the symbol equation for this reaction. [1]

Fe_2O_3 + 3CO ⟶

A Fe + $3CO_2$

B 2Fe + $3CO_2$

C 3Fe + CO_2

Letter ....

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(c) Chemists have designed a wide variety of alloys for different uses.

Some alloys contain iron and carbon only, whereas others contain additional metals.

The table shows the composition and properties of some alloys.

Alloy	Composition	Properties
mild steel	iron plus 0.15 – 0.30 % carbon	malleable (easy to bend), ductile (easy to pull into wire) and soft (easy to scratch)
high carbon steel	iron plus 0.70 – 1.50 % carbon	strong, brittle and hard
cast iron	iron plus 2.00 – 5.00 % carbon	very strong, very brittle and very hard

Use only the information in the table above to answer parts (i)–(iii).

(i) Name the alloy that contains the **least** amount of carbon. [1]

....

(ii) Underline the effect of increasing the percentage (%) of carbon in these alloys. [1]

strength decreases **hardness decreases**

softness increases **brittleness increases**

(iii) Give the property of mild steel which makes it useful for making car bodies. [1]

....

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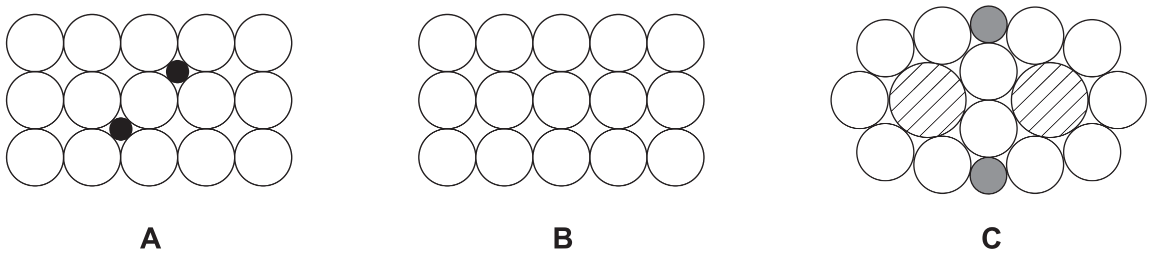
(iv) Diagrams **A**, **B** and **C** are models showing the arrangement of atoms in pure iron and in two alloys, **but not necessarily in that order**.

Choose the **letter** of the model which best represents cast iron. Give the reason for your choice. [2]



Letter ....

Reason ....

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